The Accelerating Universe as Statistical Inference: Type Ia Supernovae, Maximum Likelihood, and the Cosmological Constant, 1988–1999

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Abstract

In their Nobel Prize–winning papers, Adam Riess, Brian Schmidt, and their High-Z Supernova Search Team [Riess et al., 1998], and independently Saul Perlmutter and the Supernova Cosmology Project [Perlmutter et al., 1999], reported that the expansion of the universe is *accelerating*. Their argument had two components: a physical *model* (the Friedmann equations with a cosmological constant Λ) and a *measurement equation* relating the observable apparent magnitude of a Type Ia supernova to its redshift, with the shape of that relation controlled by two density parameters Ω_M and Ω_Λ. This note examines their reasoning from a modern statistical perspective. I describe the data assembled by both groups—the Calán/Tololo low-redshift training sample and the high-redshift supernova discoveries at $z = 0.18$–0.83—and lay out the maximum likelihood estimation problem they solved: finding $(\Omega_M, \Omega_\Lambda)$ to minimise the weighted sum of squared residuals between observed distance moduli and the theoretical prediction from the Friedmann luminosity-distance formula. Under Gaussian errors MLE coincides with weighted least squares (WLS). The curvature of the log-likelihood surface—the

*In 2011 my good friend Christopher A. Sims and I shared the Nobel Prize in economics. That good luck let us meet and become friends with winners of prizes in physics and other fields. Our conversations with the physics winners were especially interesting. Adam Riess had majored partly in economics as an undergraduate and told us about his very interesting undergraduate thesis. I have kept in contact with Adam Riess and Brian Schmidt, who have taught me the nuts and bolts of the computer programs that they used to compute and climb the likelihood function that led to their remarkable discovery.

Fisher information—determines the confidence ellipses in the $(\Omega_M, \Omega_\Lambda)$ plane; its elongation exposes the near-degeneracy of two cosmological parameters that can be resolved only by spanning a wide redshift range. A critical nuisance parameter— the combination $\mathcal{M}$ of the absolute magnitude M_B and the Hubble constant H_0—enters the measurement equation exactly as the contact electromotive force (EMF) enters Millikan's photoelectric regression: it shifts the entire Hubble diagram up or down but leaves the *shape* of $\mu(z)$ unaffected, so Ω_M and Ω_Λ are identified by the curvature of the diagram independently of $\mathcal{M}$. The MLE estimates—$\Omega_M \approx 0.28$, $\Omega_\Lambda \approx 0.72$ for a flat universe—imply a deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda \approx -0.58 < 0$, signalling acceleration. Both teams found the data incompatible with $\Omega_\Lambda = 0$ at more than 3σ. The episode is reinterpreted through the lens of modern machine learning: the Friedmann equations supplied the hypothesis class; the Phillips width–luminosity relation (analogous to Lenard's finding that frequency, not intensity, drives the photoelectric effect) provided the crucial standardization insight; and the final parameter estimation is a supervised learning problem in which the shape of the $\mu(z)$ curve is the discriminating feature.

1 Introduction

In March 1998 and January 1999, two independent teams announced that distant Type Ia supernovae are systematically *fainter* than they would be in a universe whose expansion is decelerating. The simplest explanation: the universe's expansion is speeding up, driven by a positive cosmological constant Λ or some analogous "dark energy." For this discovery, Saul Perlmutter received one half of the 2011 Nobel Prize in Physics, and Adam Riess and Brian Schmidt shared the other half.

A twenty-first-century statistician confronted with tables of (redshift, apparent magnitude) pairs would recognise the problem as a weighted nonlinear regression with two scientifically meaningful parameters and a nuisance intercept. The Friedmann equations with a cosmological constant were well established before the data were collected. The discovery consisted in determining the sign and magnitude of Ω_Λ.

The story has five features of interest to a statistician.

1. **Model specification: the hypothesis class from the Friedmann equations.** General relativity predicts the luminosity-distance–redshift relation $d_L(z; \Omega_M, \Omega_\Lambda)$

exactly. The functional form is fixed, and the estimation problem reduces to two dimensionless density parameters.

2. **Standardizing a noisy standard candle.** The peak B-band luminosities of raw Type Ia supernovae scatter by ≈ 0.4 magnitudes. Phillips [1993] found that the rate of post-peak decline Δm_{15} predicts peak luminosity, which reduces the dispersion to ≈ 0.17 mag.

3. **The nuisance parameter $\mathcal{M}$.** The combination of absolute magnitude M_B and Hubble constant H_0 shifts the entire Hubble diagram vertically. The observations do not identify $\mathcal{M}$ on its own; it is profiled out of the likelihood. The relative distance moduli of high-z and low-z supernovae identify the shape of $\mu(z)$ and with it Ω_M and Ω_Λ.

4. **Independent replication with different methods.** The High-Z Supernova Search Team used the MLCS (Multi-Light-Curve Shape) standardization method of Riess et al. [1996]; the Supernova Cosmology Project used the stretch-factor method of Perlmutter et al. [1997]. The two groups analysed different supernovae and reached the same conclusion to within their measurement uncertainties.

5. **Reluctance to accept the model.** Many physicists doubted $\Omega_\Lambda > 0$ because the prevailing theory held that the vacuum energy density should be negligible [Weinberg, 1989].

Both teams had roughly 40–60 high-quality supernovae, a meagre dataset by modern standards. The Friedmann equations specified the regression function up to three free parameters and so collapsed a high-dimensional function-approximation task into a low-dimensional fitting problem that the available data could settle.

Sections 2–4 set the observational and theoretical stage. Section 5 presents the WLS/MLE procedure. Section 6 develops the Fisher information and confidence ellipses. Section 7 discusses the $\mathcal{M}$ identification problem. Section 8 presents the main results and the inference of acceleration. Section 9 compares the two teams. Section 10 describes the scepticism that greeted the result and the robustness tests that answered it. Sections 11–12 reinterpret the episode in the language of machine learning. Section 13 concludes.

2 Observational Background, 1929–1994

2.1 Hubble's Law and the Standard Model of Cosmology

In 1929 Edwin Hubble [1929] reported that the recession velocities v of nearby "extra-galactic nebulae" satisfy

$$v = H_0\, d, \tag{1}$$

where d is the distance and H_0 the Hubble constant. Hubble estimated $H_0 \approx 500\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; subsequent Cepheid-variable distance rescaling reduced the accepted value to $H_0 \approx 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Alexander Friedmann had already shown in 1922 and 1924 [Friedmann, 1922, 1924] that expanding cosmological solutions follow from Einstein's field equations; Hubble's measurement confirmed that we live in one of them.

In the Friedmann model, the dynamical variable is the scale factor $a(t)$, normalised so that $a(t_0) = 1$ today. The recession redshift z of a source satisfies $1 + z = 1/a(t_{\mathrm{emit}})$. Three dimensionless density parameters characterise the matter-energy content:

$$\Omega_M = \frac{8\pi G \rho_{M,0}}{3H_0^2}, \qquad \text{(non-relativistic matter)} \tag{2}$$

$$\Omega_\Lambda = \frac{\Lambda}{3H_0^2}, \qquad \text{(cosmological constant / dark energy)} \tag{3}$$

$$\Omega_k = \frac{-k}{a_0^2 H_0^2}, \qquad \text{(spatial curvature)} \tag{4}$$

with the constraint $\Omega_M + \Omega_\Lambda + \Omega_k = 1$. A flat universe has $\Omega_k = 0$, i.e., $\Omega_M + \Omega_\Lambda = 1$.

Einstein had introduced $\Lambda > 0$ in 1917 [1917] to obtain a static universe, then abandoned it after Hubble established expansion. The two teams' discovery that $\Omega_\Lambda \approx 0.72$ rehabilitated the cosmological constant as an observational reality.

2.2 Type Ia Supernovae as Standard Candles

A *standard candle* is an astronomical object of known intrinsic luminosity. If absolute luminosity L is known, the observed flux $F = L/(4\pi d_L^2)$ gives the luminosity distance d_L directly. Comparing d_L with the redshift z maps the distance-redshift relation and thereby constrains the Friedmann parameters.

Type Ia supernovae are thermonuclear explosions of carbon-oxygen white dwarfs that approach the Chandrasekhar mass limit $M_{\mathrm{Ch}} \approx 1.4\,M_\odot$ through accretion or binary-star merger. Because the exploding mass is near a universal critical value, the peak luminosity is nearly the same across all events [Filippenko, 1997]. Before standardization, the scatter in peak B-band absolute magnitude is $\sigma_{M_B} \approx 0.4$ mag, corresponding to a $\approx 40\%$ distance uncertainty—too large for cosmological measurements.

2.3 The Width-Luminosity Relation: The Key Standardization Insight

Phillips [1993] established a tight empirical relation between the peak luminosity of a Type Ia supernova and the speed at which its light curve declines after maximum light. The parameter $\Delta m_{15}(B)$ measures the drop in B-band magnitude during the 15 days after peak: brighter (more energetic) supernovae have slower, broader light curves (Δm_{15} smaller), while fainter events decline more rapidly. Quantitatively, the relation is approximately

$$M_B^{\mathrm{peak}} \approx M_B^{\mathrm{ref}} + \alpha\big(\Delta m_{15} - 1.1\big), \quad \alpha \approx 0.75 \text{ mag}, \tag{5}$$

where M_B^{ref} is the absolute magnitude of a "normal" event with $\Delta m_{15} = 1.1$. After applying this correction, the intrinsic dispersion falls to $\sigma_{M_B} \approx 0.17$ mag.

The Phillips relation is the supernova analogue of Lenard's discovery that the photoelectric stopping potential depends on frequency, not intensity. Lenard's finding let Einstein specify the input variable ν; the Phillips relation identifies Δm_{15} as the feature that standardizes Type Ia peak magnitudes. Without it the scatter masks the cosmological signal. With it the $\mu(z)$ residuals distinguish a decelerating from an accelerating universe.

2.4 The Two Teams

The **Supernova Cosmology Project** (SCP), led by Saul Perlmutter at Lawrence Berkeley National Laboratory, began systematic high-z supernova searches in the late 1980s and developed a systematic discovery pipeline using ground-based telescopes (Isaac Newton Telescope, Cerro Tololo Inter-American Observatory, Keck). Their strategy, described in Goobar and Perlmutter [1995], exploited "batch mode" scheduling: supernova candidates were discovered in a first observing run, then followed up in a *prescheduled* second run three weeks later, guaranteeing that the events were still near peak brightness when follow-up spectroscopy was obtained. By 1997 they had reported results for seven high-z supernovae [Perlmutter et al., 1997]; their definitive 42-supernova paper appeared in 1999 [Perlmutter et al., 1999].

The **High-Z Supernova Search Team** (HZT), assembled around Brian Schmidt at the Mount Stromlo Observatory and Robert Kirshner at Harvard, made complementary use of the Hubble Space Telescope and larger ground-based facilities to obtain high-quality light curves and spectra. Adam Riess developed the MLCS analysis pipeline for the team's data [Riess et al., 1996]. A first report on four HST supernovae [Garnavich et al., 1998] was followed by the definitive 16-plus-34 SN paper of Riess et al. [1998].

3 The Cosmological Model and the Measurement Equation

3.1 The Friedmann Equation

The first Friedmann equation with matter density ρ_M, pressure-free dust, and cosmological constant Λ is

$$H^2(z) \equiv \left(\frac{\dot{a}}{a}\right)^2 = H_0^2 \Big[\Omega_M (1+z)^3 + \Omega_k (1+z)^2 + \Omega_\Lambda\Big] \equiv H_0^2\, E^2(z), \tag{6}$$

where z is evaluated on the past light cone and the present-day normalisation $E(0) = 1$ implies $\Omega_M + \Omega_k + \Omega_\Lambda = 1$. The dimensionless Hubble function $E(z)$ gives the expansion rate at redshift z in units of H_0. For $\Omega_\Lambda = 0$ and matter-only content, $E(z) = (1+z)^{3/2}$ (the Einstein-de Sitter solution); for the best-fit flat ΛCDM model, $E(z)$ passes through intermediate values between these extremes.

3.2 Luminosity Distance

The luminosity distance to a source at redshift z is

$$d_L(z; \Omega_M, \Omega_\Lambda) = \frac{c(1+z)}{H_0 \sqrt{|\Omega_k|}}\, f\left(\sqrt{|\Omega_k|} \int_0^z \frac{dz'}{E(z')}\right), \tag{7}$$

where

$$f(x) = \begin{cases} \sin x & \Omega_k < 0 \text{ (closed)}, \\ x & \Omega_k = 0 \text{ (flat)}, \\ \sinh x & \Omega_k > 0 \text{ (open)}. \end{cases} \tag{8}$$

For the flat case of primary interest ($\Omega_k = 0$, $\Omega_M + \Omega_\Lambda = 1$), equation (7) reduces to the elegant form

$$\boxed{d_L(z; \Omega_M) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_M(1+z')^3 + (1-\Omega_M)}}.} \tag{9}$$

Here $\Omega_\Lambda = 1 - \Omega_M$ is not a free parameter once flatness is imposed. For the general (non-flat) case, Ω_M and Ω_Λ are estimated jointly without the flatness constraint.

3.3 The Distance Modulus and Measurement Equation

The *distance modulus* is defined as

$$\mu \equiv m_B - M_B = 5 \log_{10}\left(\frac{d_L}{10 \text{ pc}}\right), \tag{10}$$

where m_B is the observed peak apparent magnitude (standardized for light-curve shape) and M_B is the intrinsic absolute magnitude. Since 1 Mpc $= 10^6$ pc, this becomes

$$\mu = 5 \log_{10}\left(\frac{d_L}{\text{Mpc}}\right) + 25. \tag{11}$$

The theoretical distance modulus predicted by the cosmological model is therefore

$$\mu^{\text{th}}(z; \Omega_M, \Omega_\Lambda, H_0) = 5 \log_{10}\left(\frac{d_L(z; \Omega_M, \Omega_\Lambda)}{\text{Mpc}}\right) + 25. \tag{12}$$

The full measurement equation—the fundamental regression of the entire analysis—is

$$\boxed{m_{B,i} = M_B + \mu^{\text{th}}(z_i; \Omega_M, \Omega_\Lambda, H_0) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2),} \tag{13}$$

where the sum of squared errors will define the likelihood and the free parameters to be estimated are M_B, H_0, Ω_M, and Ω_Λ.

3.4 The Nuisance Parameter $\mathcal{M}$

Expanding equation (12) reveals that H_0 and M_B enter only through a single combination. Write

$$d_L(z;\Omega_M,\Omega_\Lambda,H_0)=\frac{c}{H_0}\,\tilde{d}_L(z;\Omega_M,\Omega_\Lambda), \tag{14}$$

where $\tilde{d}_L$ is the *dimensionless* luminosity distance that depends only on the density parameters. Then

$$\begin{aligned}\mu^{\rm th}&=5\log_{10}\left(\frac{c}{H_0}\,\tilde{d}_L(z;\Omega_M,\Omega_\Lambda)\right)+25\\&=5\log_{10}\tilde{d}_L(z;\Omega_M,\Omega_\Lambda)+5\log_{10}\left(\frac{c}{H_0\,\mathrm{Mpc}}\right)+25,\end{aligned} \tag{15}$$

so the measurement equation (13) becomes

$$m_{B,i}=\underbrace{M_B+5\log_{10}\left(\frac{c}{H_0\,\mathrm{Mpc}}\right)+25}_{\mathcal{M}}+5\log_{10}\tilde{d}_L(z_i;\Omega_M,\Omega_\Lambda)+\varepsilon_i. \tag{16}$$

The scalar $\mathcal{M}$ shifts the Hubble diagram vertically; the term $5\log_{10}\tilde{d}_L(z;\Omega_M,\Omega_\Lambda)$ determines its shape and does not depend on $\mathcal{M}$.

- $\mathcal{M}$ is not identified by the shape of the Hubble diagram. Any vertical shift of the entire m_B sequence can be absorbed into $\mathcal{M}$.
- Ω_M and Ω_Λ *are* identified by the *curvature* of $\mu(z)$: the way the slope of the m_B–z diagram changes between low-z and high-z supernovae.

Section 7 compares this identification structure with Millikan's contact-EMF problem.

The estimating model has three effective parameters:

$$\boldsymbol{\theta}=(\mathcal{M},\,\Omega_M,\,\Omega_\Lambda), \tag{17}$$

plus an intrinsic scatter $\sigma_{\rm int}$. For a flat universe the constraint $\Omega_M+\Omega_\Lambda=1$ reduces the shape parameters to the single free parameter Ω_M.

4 The Historical Data

4.1 The Low-Redshift Calibration Sample

Measuring $(\Omega_M, \Omega_\Lambda)$ requires supernovae spanning a wide enough redshift range that the *shape* of the $\mu(z)$ curve—which changes between different cosmological models—is statistically distinguishable. Both teams anchored their analyses with a common low-redshift ($z \lesssim 0.1$) training set: the Calán/Tololo Supernova Survey of Hamuy et al. [1996], which provided 29 carefully observed Type Ia supernovae in BVI bands. Table 1 lists a representative subset.

Table 1: Representative Type Ia supernovae from the Calán/Tololo low-redshift sample [Hamuy et al., 1996]. The effective B-band peak magnitude m_B^{eff} incorporates the light-curve shape correction of equation (5). These events serve as the "training data" in section 11: they calibrate the absolute magnitude M_B for use in the high-z sample.

Supernova	z	m_B^{eff} (mag)	σ (mag)
SN 1990O	0.030	16.33	0.07
SN 1992P	0.026	16.08	0.07
SN 1992ae	0.075	18.44	0.09
SN 1992al	0.014	14.60	0.06
SN 1992bc	0.020	15.18	0.06
SN 1992bh	0.045	17.42	0.08
SN 1992bl	0.043	17.29	0.08
SN 1992bo	0.018	15.95	0.06
SN 1993ag	0.050	17.68	0.09
SN 1993O	0.052	17.72	0.08

At low redshifts, $d_L \approx cz/H_0$ (Hubble's law), so $\mu \approx 5\log_{10}(cz/H_0 \cdot \text{Mpc}) + 25$; the shape parameters $(\Omega_M, \Omega_\Lambda)$ enter only at order z^2 and are not measurable from $z \lesssim 0.1$ data alone. The low-z sample thus constrains $\mathcal{M}$ (through Hubble's law), not $(\Omega_M, \Omega_\Lambda)$.

4.2 The High-Redshift Samples

Both teams spent several years assembling high-z supernova samples using ground-based telescopes supplemented by Hubble Space Telescope photometry and Keck spectroscopy.

Perlmutter et al. (1999). The SCP's "Fit C" sample comprises 18 high-quality high-z events; their full "Fit C+D" sample extends to 42 supernovae at $z = 0.18$–0.83.

Table 2 lists a representative subset, showing the effective apparent magnitudes after K-correction (accounting for the redshift of the SN Ia spectral energy distribution relative to the B-band filter) and stretch-factor standardization.

Table 2: Representative high-redshift Type Ia supernovae from the Supernova Cosmology Project [Perlmutter et al., 1999]. Effective B-band magnitudes m_B^{eff} after stretch-factor and K-correction. The photometric uncertainty σ includes contributions from photon noise, host-galaxy subtraction, K-correction uncertainty, and intrinsic scatter $\sigma_{\text{int}} \approx$ 0.15 mag. Values are approximately reproduced from Table 1 of Perlmutter et al. [1999].

Supernova	z	m_B^{eff} (mag)	σ (mag)
SN 1994am	0.372	22.46	0.21
SN 1994H	0.374	22.17	0.15
SN 1994an	0.378	22.92	0.18
SN 1995az	0.450	22.79	0.20
SN 1994al	0.420	22.45	0.22
SN 1994G	0.425	22.55	0.17
SN 1994F	0.354	22.24	0.19
SN 1995ba	0.388	22.65	0.17
SN 1992bi	0.458	22.92	0.16
SN 1993O	0.583	23.57	0.13
SN 1996cg	0.490	23.38	0.18
SN 1997ck	0.830	24.23	0.23

Riess et al. (1998). The HZT's primary dataset comprised 16 high-z supernovae ($z = 0.16$–0.62), supplemented by 34 low-z events compiled from the literature. The MLCS method was applied to multi-band ($BVRI$) light curves, providing simultaneous estimates of the shape parameter Δ, the colour excess $E(B-V)$ (dust reddening), and the peak magnitude. A subset is shown in Table 3.

4.3 The Standardization Pipeline

Both teams corrected the raw observed magnitudes for four effects before entering them into the cosmological likelihood:

1. **K-correction.** At $z \sim 0.5$, the observer-frame I-band covers rest-frame B-band, while the observer-frame B-band covers rest-frame UV. A wavelength-dependent correction transforms the measured filter magnitude to the rest-frame B-band magnitude that anchors the low-z calibration. The SCP used templates of Kim

Table 3: Selected high-redshift supernovae from the High-Z Supernova Search Team [Riess et al., 1998]. Corrected distance moduli μ_{MLCS} are computed from the full MLCS fit to the multi-band light curves and incorporate corrections for light-curve shape (Δ) and dust reddening (A_B). Values approximately reproduced from Riess et al. [1998], Table 5.

Supernova	z	μ_{MLCS}	σ_μ	Notes
SN 1997ap	0.830	44.02	0.45	HST photometry
SN 1996cl	0.828	44.41	0.31	"golden"
SN 1997ce	0.440	42.53	0.33	"golden"
SN 1997cj	0.500	42.91	0.24	"golden"
SN 1997ck	0.970	44.30	0.51	high-z outlier
SN 1995ao	0.300	41.08	0.36	
SN 1995ap	0.230	40.36	0.26	
SN 1996cf	0.570	43.16	0.42	
SN 1996bj	0.574	43.67	0.26	

et al. [1996]; the HZT used spectrophotometric templates from Nugent et al. [2002].

2. **Light-curve shape correction.** The SCP employed a *stretch factor* s (time-axis scaling): each light curve is remapped to the template by stretching the time axis by s, then the peak magnitude is corrected by $\alpha_{\mathrm{SCP}}(s-1)$ with $\alpha_{\mathrm{SCP}} \approx 1.35$ mag. The HZT's MLCS method fit multi-band templates parameterised by a single luminosity offset Δ; the correction is $\eta\,\Delta$ with $\eta \approx 0.91$ mag per unit Δ.

3. **Colour/dust correction.** Both teams corrected for dust reddening in the host galaxy using the measured $B-V$ colour excess at peak light: $A_B = R_B\,E(B-V)$ with $R_B \approx 4.1$.

4. **Peculiar-velocity uncertainty.** Each SN at $z < 0.1$ contributes a velocity uncertainty $\sigma_v \approx 200$–$300\,\mathrm{km/s}$ from large-scale structure motions, adding an effective distance- modulus error $\sigma_{v,i} = 5\sigma_v/(cz_i \ln 10)$ that diverges at low z. This is why the minimum useful redshift is $z \approx 0.01$.

After these corrections, the effective m_B^{eff} enters the cosmological likelihood as if it were the true distance modulus plus a Gaussian residual.

The total photometric uncertainty per supernova is typically

$$\sigma_{m,i}^2 = \sigma_{\text{photo,i}}^2 + \sigma_{K,i}^2 + \sigma_{v,i}^2 + \sigma_{\text{int}}^2,$$

where $\sigma_{\text{int}} \approx 0.13$–$0.17$ mag is the residual dispersion after standardization.

5 Weighted Least Squares and Maximum Likelihood Estimation

5.1 The Likelihood Function

Given the measurement model (16), in which each observation $m_{B,i}$ is drawn from a normal distribution centred on $\mathcal{M} + 5\log_{10}\tilde{d}_L(z_i;\Omega_M,\Omega_\Lambda)$, the likelihood for a sample of $n = N_{\text{low}} + N_{\text{high}}$ supernovae is

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma_i} \exp\left(-\frac{\left[m_{B,i} - \mathcal{M} - 5\log_{10}\tilde{d}_L(z_i;\Omega_M,\Omega_\Lambda)\right]^2}{2\sigma_i^2} \right), \tag{18}$$

where the total uncertainty σ_i (including intrinsic scatter) was discussed in Section 4. The log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2}\ln(2\pi) - \sum_{i=1}^{n} \ln\sigma_i - \underbrace{\frac{1}{2}\sum_{i=1}^{n} \frac{\left[m_{B,i} - \mathcal{M} - 5\log_{10}\tilde{d}_L(z_i;\Omega_M,\Omega_\Lambda)\right]^2}{\sigma_i^2}}_{\chi^2(\boldsymbol{\theta})/2}. \tag{19}$$

Maximising (19) over $\boldsymbol{\theta}$ minimises $\chi^2(\boldsymbol{\theta})$.

5.2 MLE Equals WLS Under Gaussian Errors

For fixed σ_i (i.e., ignoring the dependence of σ_{int} on $\boldsymbol{\theta}$), maximising (19) over $(\mathcal{M},\Omega_M,\Omega_\Lambda)$ requires minimising

$$\chi^2(\mathcal{M},\Omega_M,\Omega_\Lambda) = \sum_{i=1}^{n} \frac{\left[m_{B,i} - \mathcal{M} - 5\log_{10}\tilde{d}_L(z_i;\Omega_M,\Omega_\Lambda)\right]^2}{\sigma_i^2}. \tag{20}$$

Unlike the photoelectric case—where OLS had a closed-form solution— minimising

(20) over $(\Omega_M, \Omega_\Lambda)$ requires numerical optimisation because $\tilde{d}_L(z; \Omega_M, \Omega_\Lambda)$ involves a numerical integral rather than an algebraic formula. Both groups used grid search over a fine $(\Omega_M, \Omega_\Lambda)$ lattice, evaluating χ^2 at each grid point via numerical quadrature. The estimator is still the minimiser:

$$\boxed{(\hat{\mathcal{M}}_{\text{MLE}},\, \hat{\Omega}_{M,\text{MLE}},\, \hat{\Omega}_{\Lambda,\text{MLE}}) = \underset{(\mathcal{M}, \Omega_M, \Omega_\Lambda)}{\arg\min}\ \chi^2(\mathcal{M}, \Omega_M, \Omega_\Lambda).} \tag{21}$$

5.3 Profiling Out the Nuisance Parameter

$\mathcal{M}$ can be eliminated analytically at any fixed $(\Omega_M, \Omega_\Lambda)$, because it enters (20) linearly:

$$\hat{\mathcal{M}} = \frac{\displaystyle\sum_i \frac{m_{B,i} - 5\log_{10}\tilde{d}_L(z_i)}{\sigma_i^2}}{\displaystyle\sum_i \frac{1}{\sigma_i^2}}. \tag{22}$$

Substituting (22) into χ^2 gives the *profile chi-squared*

$$\tilde{\chi}^2(\Omega_M, \Omega_\Lambda) \equiv \chi^2(\hat{\mathcal{M}}, \Omega_M, \Omega_\Lambda), \tag{23}$$

which depends only on the two shape parameters and can be minimised on a two-dimensional grid. In practice both groups also marginalised or profiled over intrinsic scatter σ_{int}.

5.4 Predicted Distance Moduli for Representative Cosmological Models

Table 4 computes the theoretical distance modulus $\mu^{\text{th}}(z; \Omega_M, \Omega_\Lambda, H_0)$ for four cosmological models at representative redshifts, using $H_0 = 70\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$. The model predictions diverge most strongly at $z \approx 0.3$–0.8.

At $z \approx 0.5$, the flat ΛCDM model $(\Omega_M = 0.3, \Omega_\Lambda = 0.7)$ predicts supernovae that are ≈ 0.21 mag fainter (larger μ) than the open matter-only model $(\Omega_M = 0.3, \Omega_\Lambda = 0)$. Since the standardized Type Ia dispersion is $\sigma_{\text{int}} \approx 0.15$ mag, every event at $z \approx 0.5$ contributes about 1.4σ of discriminatory signal. With 10–15 well-measured supernovae near $z = 0.5$, the combined signal-to-noise is ≈ 3–4σ, which matches the announcements

Table 4: Theoretical distance modulus $\mu = 5\log_{10}(d_L/\mathrm{Mpc}) + 25$ for $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ under four cosmological models. The luminosity distances are computed by numerical integration of equation (7). $\Delta\mu_\Lambda$ is the excess distance modulus of the flat-ΛCDM model relative to the open matter-only model; positive values mean the flat-ΛCDM universe predicts *fainter* supernovae.

	$\mu^{\rm th}$ (mag)				
z	Flat ΛCDM (0.3, 0.7)	EdS (1.0, 0.0)	Open (0.3, 0.0)	Flat matter (0.3, 0.0, flat)	$\Delta\mu_\Lambda$ (Λ − open)
0.10	38.44	38.38	38.42	38.38	+0.02
0.25	40.51	40.27	40.45	40.27	+0.06
0.50	42.26	41.86	42.05	41.86	+0.21
0.75	43.33	42.82	43.06	42.82	+0.27
1.00	44.10	43.53	43.83	43.53	+0.27

of both groups.

The excess faintness found in both the Riess et al. and Perlmutter et al. analyses, $\Delta\mu_{\rm obs} \approx +0.25$ mag at $z \approx 0.5$ (see Section 8), sits between the predictions for (0.3, 0.7) and (0.3, 0.0), comfortably rejecting the latter at more than 3σ.

6 Fisher Information and Confidence Ellipses

6.1 The Fisher Information Matrix

The precision with which $(\Omega_M, \Omega_\Lambda)$ can be estimated is governed by the Fisher information matrix $\mathcal{I}(\boldsymbol{\theta})$. For the model (16) with known σ_i:

$$\mathcal{I}_{jk}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{1}{\sigma_i^2} \frac{\partial\mu^{\rm th}(z_i;\boldsymbol{\theta})}{\partial\theta_j} \frac{\partial\mu^{\rm th}(z_i;\boldsymbol{\theta})}{\partial\theta_k}, \tag{24}$$

where the sum runs over all supernovae. The Cramér-Rao bound states that any unbiased estimator $\tilde{\boldsymbol{\theta}}$ satisfies $\mathrm{Var}(\tilde{\theta}_j) \geq [\mathcal{I}^{-1}]_{jj}$.

For the shape parameters $(\Omega_M, \Omega_\Lambda)$ the relevant derivatives are

$$\frac{\partial\mu}{\partial\Omega_M} = \frac{5}{\ln 10}\frac{1}{\tilde{d}_L}\frac{\partial\tilde{d}_L}{\partial\Omega_M}, \qquad \frac{\partial\mu}{\partial\Omega_\Lambda} = \frac{5}{\ln 10}\frac{1}{\tilde{d}_L}\frac{\partial\tilde{d}_L}{\partial\Omega_\Lambda}. \tag{25}$$

These are computed numerically. The parameter Ω_M slows the expansion and reduces μ; the parameter Ω_Λ accelerates the expansion and raises μ. At moderate redshifts $z \sim 0.5$,

$$\frac{\partial\mu/\partial\Omega_M}{\partial\mu/\partial\Omega_\Lambda} \approx -\frac{3}{4}, \tag{26}$$

so the two parameters are nearly degenerate: increasing Ω_M by 0.75 units and simultaneously increasing Ω_Λ by 1 unit leaves $\mu(z \approx 0.5)$ approximately constant.

The degeneracy direction in the $(\Omega_M, \Omega_\Lambda)$ plane satisfies

$$0.8\,\Omega_M - 0.6\,\Omega_\Lambda \approx \text{const}, \tag{27}$$

a result derived analytically by Goobar and Perlmutter [1995]. The confidence ellipses are elongated *along* this line; the *perpendicular* direction (the "constraint direction")

$$0.6\,\Omega_M + 0.8\,\Omega_\Lambda \approx \text{const}$$

is well constrained by the data.

6.2 The Value of Wide Redshift Coverage

Goobar and Perlmutter [1995] showed that breaking the degeneracy (27) requires supernovae at two well-separated redshifts. A sample restricted to $z \approx 0.5$ traces a near-degenerate path in the $(\Omega_M, \Omega_\Lambda)$ plane. Adding $z \approx 0.8$–1.0 events rotates the constraint direction.

The SCP's highest-redshift event, SN 1997ck at $z = 0.83$, was particularly important: it lay nearly orthogonal to the degeneracy direction and alone contributed significant constraint-direction information. The HZT's HST-observed events at $z > 0.8$ served the same role.

6.3 Numerical Illustration of the Confidence Ellipses

Perlmutter et al. [1999] reported their confidence contours directly as $\Delta\chi^2 = \tilde{\chi}^2 - \tilde{\chi}^2_{\min}$ contours. The 68% and 95% confidence ellipses for one free parameter correspond to $\Delta\chi^2 = 1$ and 4, respectively; for two parameters, to $\Delta\chi^2 = 2.30$ and 5.99. In the published confidence plots (Figure 2 of Perlmutter et al. 1999; Figure 7 of Riess et al.

1998) the best-fit region lies in the quadrant $\Omega_\Lambda > 0$, more than 3σ from the line $\Omega_\Lambda = 0$ in the constraint direction.

7 The $\mathcal{M}$ Identification Problem

7.1 The Contact-EMF Analogy

In Millikan's photoelectric regression, the contact electromotive force v_c between the photoemitting metal and the collector entered the stopping-potential equation as a pure intercept:

$$V_{0i} = \underbrace{-W/e + v_c}_{\alpha} + (h/e)\,\nu_i + \varepsilon_i.$$

The relevant physics—the slope h/e—was identified from the *gradient* of the V_0–ν relationship, not its level. Millikan controlled v_c by measuring it continuously with a Kelvin double-balance; because it remained constant during a single experimental run, it shifted all observations equally and did not bias $\hat{\beta}$.

In the supernova problem, $\mathcal{M} = M_B + 5\log_{10}(c/H_0\,\mathrm{Mpc}) + 25$ plays the role of v_c. The relevant physics—$(\Omega_M, \Omega_\Lambda)$—is identified from the *shape* (curvature) of the m_B–z relationship, not its level. Neither M_B nor H_0 can be determined independently from the Hubble diagram because any value of H_0 can be compensated exactly by a corresponding shift in M_B. The shape parameters are not affected by this degeneracy.

7.2 Constraining $\mathcal{M}$ from the Low-z Sample

Both teams' solution was identical in spirit to Millikan's: constrain the nuisance parameter from an independent source. At low z, $d_L \approx cz/H_0$, so $m_{B,i} \approx \mathcal{M} + 5\log_{10} z_i + C$ (a constant). The Calán/Tololo low-z sample calibrates $\mathcal{M}$ through Hubble's law (1), with uncertainties from peculiar velocities averaged away over ≈ 18 events. Once $\mathcal{M}$ is anchored by the low-z data, the high-z data constrain $(\Omega_M, \Omega_\Lambda)$ through the curvature of $\mu(z)$.

7.3 Separate Identification of H_0 and M_B

Even with the low-z anchor, H_0 and M_B are individually identified only if M_B is calibrated by an external distance ladder. The primary constraint came from Cepheid variable distances measured with the Hubble Space Telescope Key Project, which established $H_0 = 72 \pm 8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (circa 1998). Given H_0, the Calán/Tololo low-z sample then determines $M_B \approx -19.3$ mag, confirming that Type Ia supernovae are among the most luminous transient phenomena in the universe.

M_B, equivalently H_0, is a nuisance parameter for the cosmological inference. Its uncertainty propagates into $\mathcal{M}$ but not into $(\Omega_M, \Omega_\Lambda)$. A 10% error in H_0 shifts the Hubble diagram by ≈ 0.22 mag and leaves the best-fit Ω_M and Ω_Λ unchanged.

8 Results: The Deceleration Parameter and Cosmic Acceleration

8.1 Best-Fit Parameters

Table 5 summarises the principal parameter estimates from both groups, using their primary standardization methods.

Table 5: Summary of cosmological parameter estimates from the two Nobel-Prize groups. All results use the respective team's primary light-curve standardization method. "Flat" imposes $\Omega_M + \Omega_\Lambda = 1$. Statistical uncertainties only unless stated. The deceleration parameter is $q_0 = \Omega_M/2 - \Omega_\Lambda$.

Analysis	$\hat{\Omega}_M$	$\hat{\Omega}_\Lambda$	$\hat{q}_0$	Ref.
Perlmutter et al. (SCP), flat	0.28 ± 0.09	0.72 ± 0.09	-0.58	Perlmutter et al. [1999]
Perlmutter et al. (SCP), free	0.28 ± 0.09	0.72 ± 0.10	-0.58	Perlmutter et al. [1999]
Riess et al. (HZT), MLCS, flat	0.28 ± 0.10	0.72 ± 0.10	-0.58	Riess et al. [1998]
Riess et al. (HZT), template	0.20 ± 0.10	0.80 ± 0.10	-0.70	Riess et al. [1998]
Planck 2018 (CMB+BAO, flat)	0.315 ± 0.007	0.685 ± 0.007	-0.528	Planck Collaboration et al. [2020]

Both groups found $\hat{\Omega}_\Lambda \approx 0.7$ from independent datasets and different standardization pipelines. Section 9 compares them.

8.2 The Deceleration Parameter

From the second Friedmann equation,

$$q_0 \equiv -\frac{\ddot{a}\,a}{\dot{a}^2}\bigg|_{t=t_0} = \frac{\Omega_M}{2} - \Omega_\Lambda. \tag{28}$$

For the best-fit flat model ($\hat{\Omega}_M = 0.28$, $\hat{\Omega}_\Lambda = 0.72$):

$$\hat{q}_0 = \frac{0.28}{2} - 0.72 = 0.14 - 0.72 = \mathbf{-0.58}. \tag{29}$$

The universe is accelerating. The transition from deceleration to acceleration occurred at the redshift z_t satisfying $\Omega_M(1+z_t)^3 = 2\Omega_\Lambda$, which gives

$$z_t = \left(\frac{2\Omega_\Lambda}{\Omega_M}\right)^{1/3} - 1 \approx \left(\frac{1.44}{0.28}\right)^{1/3} - 1 \approx (5.14)^{1/3} - 1 \approx 0.73. \tag{30}$$

At $z \lesssim 0.73$, dark energy dominates and expansion accelerates; at $z \gtrsim 0.73$, matter dominates and expansion decelerates. Riess et al. [2004] subsequently discovered high-z supernovae at $z > 1$ that showed the predicted past deceleration, placing the transition at $z_t \approx 0.46$ (accounting for parameter revisions from later CMB data).

8.3 The Residual Hubble Diagram

The residual Hubble diagram is the difference between observed distance moduli and those predicted by the best-fit matter-only open model. Define

$$\Delta\mu_{\mathrm{obs,i}} \equiv \mu_{\mathrm{obs,i}} - \mu^{\mathrm{th}}(z_i; \Omega_M = 0.3, \Omega_\Lambda = 0). \tag{31}$$

For the best-fit flat ΛCDM model, $\Delta\mu^{\mathrm{th}}(z = 0.5) \approx +0.21$ mag (Table 4).

From both datasets, the weighted mean residual for high-z supernovae is

$$\overline{\Delta\mu}_{\mathrm{obs}}(z \approx 0.3\text{-}0.7) \approx +0.25 \pm 0.06 \text{ mag}. \tag{32}$$

This observed excess faintness is $\approx 4\sigma$ away from $\Delta\mu = 0$ (the null model with $\Omega_\Lambda = 0$), and it matches the prediction of the flat ΛCDM model to within 1σ.

The supernovae are fainter, hence further away, than expected. The universe expanded more between then and now than a decelerating universe would have.

8.4 Statistical Significance

For Perlmutter et al. (1999):

- $\chi^2_{\min}/\nu \approx 1.03$ for the best-fit flat model (42 SNe, 2 free parameters after profiling $\mathcal{M}$, so $\nu = 39$ degrees of freedom).
- The no-Λ hypothesis ($\Omega_\Lambda = 0$) gives $\Delta\chi^2 \approx 10.8$ (1 d.o.f.), corresponding to $p < 0.001$, i.e., rejected at 3.3σ.
- Profiling over both parameters for the constraint $q_0 \geq 0$ (decelerating universe) gives rejection at $> 99.9\%$ confidence.

For Riess et al. (1998), the MLCS method gave:

- $q_0 < 0$ at 3.0σ confidence using the MLCS method ("golden" + all-high-z sample).
- $q_0 < 0$ at 2.8σ using the template-fitting method.
- The two methods agreed across independent analysis pipelines, so the result is not an artifact of the MLCS standardization.

9 Comparing the Two Teams

Table 6 summarises the differences and agreements between the two groups.

The teams used different telescopes, observing strategies, standardization algorithms, and statistical analysis packages. Their best-fit $\hat{\Omega}_M$ agree to two significant figures. Millikan's sodium and lithium slopes agreed to 0.2%; the two supernova teams' $\hat{\Omega}_M$ agree more closely than that.

10 Epistemic Scepticism and Robustness Tests

10.1 The Initial Reaction

Millikan's 1916 confirmation of Einstein's photoelectric equation was greeted with scepticism about the physical interpretation, even as the equation itself was accepted. The supernova result of 1998–1999 provoked a structurally similar reaction: many

Table 6: Comparison of the two groups' datasets, methods, and principal results. Both teams reached the same conclusion from independent data and different analysis pipelines.

Feature	High-Z Team [Riess et al., 1998]	Supernova Cosmology Project [Perlmutter et al., 1999]
High-z sample size	16 SNe at $z = 0.16$–0.62	42 SNe at $z = 0.18$–0.83
Low-z sample	34 SNe (literature)	18 SNe (Calán/Tololo)
Standardization	MLCS (multi-colour light curve templates; Δ parameter)	Stretch factor (s; time-axis scaling of single-colour template)
Colour correction	$E(B-V)$ from multi-colour light curve fit	$E(B-V)$ from $B-V$ at peak
K-correction	Nugent et al. [2002] spectral templates	Kim et al. [1996] colour-matching
Intrinsic scatter	$\sigma_{\rm int} = 0.14$ mag	$\sigma_{\rm int} = 0.13$ mag
Primary result (flat)	$\hat{\Omega}_M = 0.28 \pm 0.10$	$\hat{\Omega}_M = 0.28 \pm 0.09$
q_0 significance	$q_0 < 0$ at 3.0σ	$q_0 < 0$ at $> 3\sigma$

physicists accepted the statistical result but questioned whether it truly implied $\Omega_\Lambda > 0$, proposing instead that systematic errors mimicked the acceleration signal.

The principal alternative explanations advanced were:

1. **Dust dimming.** If a smooth screen of "grey" (wavelength- independent) dust occupied intergalactic space, it would absorb SN light at all redshifts preferentially at high z, making distant SNe appear fainter without the colour reddening that normally betrays dust. Both teams addressed this by checking the colour residuals of their SNe: if grey dust were responsible, the $B - V$ colour excess would scale with the magnitude offset, but no such correlation was found at the level required to explain the signal.

2. **Luminosity evolution.** If Type Ia supernovae at $z \sim 0.5$ were intrinsically ≈ 0.25 mag fainter than their low-z counterparts (due to differences in progenitor age, composition, or environment), the signal could be entirely spurious. The teams tested this by comparing the Phillips Δm_{15} distributions and colour distributions of high-z and low-z SNe; no systematic difference was found. A subsequent critical test came from Riess et al. [2004]: if the signal were due to luminosity evolution

(which would affect all SNe equally, early and late), then supernovae at $z > 1$, deep in the matter-dominated epoch, should also appear faint. Instead, Riess et al. found that SNe at $z > 1$ appeared *brighter* than the ΛCDM prediction, consistent with the predicted past *deceleration* of the universe before dark energy took over.

3. **Malmquist bias.** At high z, magnitude-limited surveys preferentially detect intrinsically bright events, biasing $\hat{m}_B$ brighter. Both teams corrected for this with detailed simulations; the bias was found to be negligible compared to the acceleration signal for their survey depths.

The acceleration signal survived all of these checks.

10.2 CMB and Large-Scale Structure Evidence

Independent corroboration came from the cosmic microwave background. The BOOMERANG balloon experiment [2000] detected the first acoustic peak of the CMB temperature power spectrum and confirmed that the total energy density satisfies $\Omega_{\text{tot}} \approx 1$ (flat universe). Combined with Hubble-constant and large-scale-structure constraints that independently fix $\Omega_M \approx 0.3$, this independently implies $\Omega_\Lambda \approx 0.7$—consistent with the supernova result at the 1% level.

By 2020 the Planck satellite's CMB measurements [Planck Collaboration et al., 2020] had raised the precision on Ω_Λ to the sub-percent level, confirming the 1998–1999 supernova estimates with more than 100σ evidence against $\Omega_\Lambda = 0$.

11 The Physicists as Machine Learners

11.1 Overview

The vocabulary of modern machine learning—supervised learning, hypothesis class, loss function, inductive bias, feature engineering, transfer learning—did not circulate in the astrophysics community of the 1990s. The two teams' reasoning maps onto it as Einstein's 1905 reasoning does. This section states the correspondence.

11.2 The Hypothesis Class

The Friedmann equations and the definition of luminosity distance fix the hypothesis class for the supernova problem:

$$\mathcal{H} = \left\{ m_B \mapsto \mathcal{M} + 5 \log_{10} \tilde{d}_L(z; \Omega_M, \Omega_\Lambda) \;\middle|\; \mathcal{M} \in \mathbb{R},\ \Omega_M \geq 0,\ \Omega_\Lambda \in \mathbb{R} \right\}. \tag{33}$$

This is a three-parameter family of smooth curves on the $(z,\, m_B)$ plane. Without the Friedmann equations the space of possible $m_B(z)$ functions is infinite-dimensional; with them it collapses to the three real numbers $(\mathcal{M}, \Omega_M, \Omega_\Lambda)$.

The theoretical restriction to $\mathcal{H}$ was not arbitrary. It follows deductively from three independent bodies of physics:

1. General relativity (the geodesic equation in an expanding spacetime);
2. The assumption of spatial homogeneity and isotropy (the cosmological principle);
3. The assumption that pressure-free matter obeys $\rho_M \propto a^{-3}$ and the cosmological constant is truly constant.

Each assumption had been tested independently long before the supernova search. The Friedmann hypothesis class was well-motivated, not curve-fitted.

11.3 Feature Engineering: The Phillips Width–Luminosity Relation

Raw photometry of Type Ia supernovae involves hundreds of measurements at different epochs and wavelengths. Phillips [1993] found that the post-peak decline rate Δm_{15} predicts the peak luminosity, which reduces that high-dimensional representation to a one-dimensional correction. Without the Phillips relation:

- Type Ia scatter is ≈ 0.4 mag, yielding SNR ≈ 1 per supernova. With 40 high-z events, the acceleration signal (≈ 0.25 mag) would be detectable at only $\approx 1.6\sigma$: insufficient for a discovery claim.
- With the Phillips correction reducing scatter to ≈ 0.15 mag, the same 40 events achieve SNR $\approx 2.7\sigma$ per independent combination, and the full analysis reaches $> 3\sigma$.

The Phillips relation is the feature transformation that made the discovery possible. A data-driven learner given raw light curves would have to extract that representation from the data alone.

11.4 Training Data and Generalisation

The *training data* for both analyses consisted of the low-z Calán/Tololo supernovae [Hamuy et al., 1996]: the ≈ 29 events at $z = 0.01$–0.10 that calibrated the nuisance parameter $\mathcal{M}$ (and, via H_0, the absolute magnitude M_B).

The *test data*—the supernovae that contribute most of the cosmological information—were the high-z events at $z = 0.3$–0.8. Both teams applied the MLCS or stretch standardization learned from low-z supernovae unchanged to the high-z events.

- The HZT's MLCS templates and the coefficient $\eta = 0.91$ mag were learned from low-z training data and then used to correct high-z events (out-of-sample prediction).
- The SCP's stretch factor s and correction coefficient α_{SCP} were similarly trained on low-z data.

The implicit assumption—that Type Ia supernovae at $z \approx 0.5$ follow the same Phillips relation as nearby events—is a *transfer learning* assumption. Its validity was tested through the spectroscopic comparison of high-z and low-z SN Ia spectra: both teams found that the optical spectra of high-z events were, within measurement uncertainty, identical to those of low-z events of the same light-curve shape.

The cross-z universality of the shape–luminosity relation reduces the effective degrees of freedom and makes the inference tractable with a small sample.

11.5 Model Selection: Accelerating versus Decelerating Universes

At the time of both papers, four main cosmological models competed:

- **Einstein-de Sitter ($\Omega_M = 1$, $\Omega_\Lambda = 0$):** the theoretically favoured flat matter-only model.

- **Open matter-only ($\Omega_M \approx 0.3$, $\Omega_\Lambda = 0$):** consistent with galaxy cluster mass measurements.
- **Flat ΛCDM ($\Omega_M \approx 0.3$, $\Omega_\Lambda \approx 0.7$):** the dark-energy model.
- **Flat "QCDM" with dynamical dark energy ($w \neq -1$):** generalisations where Ω_Λ is replaced by a field with equation of state $w < -1/3$.

The supernova data, combined with the flat-universe CMB evidence, delivered decisive model selection in favour of flat ΛCDM: the best-fit χ^2 under the winning model was lower by $\Delta\chi^2 > 10$ than under the Einstein-de Sitter alternative. Any modern model selection criterion—AIC, BIC, Bayesian evidence—would produce overwhelming evidence for the dark-energy model.

11.6 The Deductive-Inductive Loop

Table 7 translates the supernova discovery into the language of modern machine learning.

Table 7: The hypothetico-deductive loop in the supernova episode, translated into the language of modern machine learning.

Step	Astrophysical name	ML name
Propose Friedmann $+\Lambda$	Theoretical model	Model architecture
Derive $d_L(z; \Omega_M, \Omega_\Lambda)$	Prediction from GR	Hypothesis class
Phillips relation: use Δm_{15}, not raw m_B	Standardization	Feature engineering
Calán/Tololo survey	Low-z calibration	Training data
High-z SN Ia searches	Discovery campaigns	Training-data extension
Minimise $\chi^2(\Omega_M, \Omega_\Lambda)$	Maximum likelihood	WLS training
$\Omega_\Lambda > 0$ at 3σ	Acceleration confirmed	Out-of-sample test
Same result from SCP and HZT	Cross-group replication	Zero-shot transfer

12 A Deep Neural Network on the Supernova Data

12.1 Two Hypothesis Classes Compared

The Friedmann-equations hypothesis class (33) has three free parameters after fixing σ_{int}. A deep neural network (DNN) for the regression $z \to m_B$ is a universal function approximator [Hornik et al., 1989]: a three-layer network with 16 ReLU neurons per layer has approximately

$$p = (1 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 1 + 1) = 593 \text{ parameters.}$$

The VC dimension of this architecture is $O(p \log p) \approx 5{,}700$ [Vapnik, 1998], implying that reliable generalisation requires $n \gtrsim 5{,}700$ labeled examples. The SCP had 42; the HZT had 50.

12.2 Epoch I: Before Phillips (1993)

Before the width–luminosity relation was known, the Type Ia scatter was ≈ 0.4 mag. A DNN applied to raw (z, m_B) data would face an $n = O(20)$ dataset with $\sigma_{\text{noise}} \approx 0.4$ mag.

Underdetermination. With $n = 20$ and $p = 593$, the DNN is massively overparameterised. Gradient descent finds a minimum-norm interpolant that passes through all 20 training points but is unconstrained between them [Belkin et al., 2019].

Loss of the discrimination. The cosmological signal is ≈ 0.25 mag at $z \approx 0.5$—smaller than the raw 0.4 mag scatter. Without the Phillips feature transformation, the DNN cannot separate signal from noise at all.

12.3 Epoch II: After Phillips (1993), Before Large Samples

The 18-SN Calán/Tololo dataset (1996) provides clean standardized photometry, but $n = 18$ is microscopically insufficient for a 593-parameter DNN. The specific problems are:

No residual degrees of freedom. The Friedmann model leaves $n - 3 = 15$ residual degrees of freedom for $n = 18$ observations and three free parameters—enough to estimate $\hat{\sigma}_{\text{int}}$ and compute meaningful confidence intervals. The DNN has no such decomposition: with 593 parameters and 18 data points, every weight and bias is undertrained; there are no residual degrees of freedom to estimate noise variance or standard errors.

Cannot identify the degeneracy direction. The degeneracy direction $0.8\,\Omega_M - 0.6\,\Omega_\Lambda \approx \text{const}$ (Section 6) must be identified from the *curvature* of $\mu(z)$ over a wide redshift range. A DNN trained on 18 low-z events, all bunched in the near-linear regime $z < 0.1$, has no way to estimate this curvature: the functional form is undetermined. The Friedmann model, by contrast, predicts the curvature analytically from the density parameters, requiring only a small number of well-placed high-z observations to pin it down.

12.4 Epoch III: The 1998–1999 Discovery Datasets

The SCP's 42-SN and HZT's 50-SN samples are the operative training sets for the discovery.

Overfitting in the interpolation regime. With $n = 42$ and $p = 593$, the DNN still operates in the interpolation regime: it can memorise all 42 training points exactly, but its predictions between and beyond the training redshifts are unconstrained. In particular, the DNN has no mechanism to recover the physically meaningful separation $\hat{q}_0 = -0.58$ from the overfit weights.

No Cramér-Rao bound. The cosmological analysis produced confidence ellipses from $\Delta\chi^2$ contours; the Friedmann model reduces the fit to $n - 3 = 39$ degrees of freedom. A DNN has no analogue: because its residuals are zero by construction (interpolation), there is no estimate of σ_{int}, no prediction interval, and no confidence surface over $(\Omega_M, \Omega_\Lambda)$.

No physical interpretation of weights. Even if the DNN happened to learn a curve close to $\mu(z; 0.28, 0.72)$, the weights of the network would not decompose into $(\hat{\Omega}_M, \hat{\Omega}_\Lambda)$. The connection between the learned function and the deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda$ would be opaque.

12.5 The Modern Era: Large Supernova Compilations

By the 2010s, supernova compilations reached several hundred events (e.g., Conley et al. 2011, Joint Light-Curve Analysis). In this regime a DNN with appropriate architecture and regularisation could, in principle, learn the shape of $\mu(z)$ from the data alone and converge toward the Friedmann functional form. What it still could not do:

- Identify that the learned function is $5\log_{10}\tilde{d}_L(z)$ where $\tilde{d}_L$ satisfies a specific integral from general relativity.
- Extract $(\Omega_M, \Omega_\Lambda)$ from the network weights without supplying the Friedmann equations as an external constraint.
- Relate the curvature of $\mu(z)$ to the deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda$ and hence to the question of whether the universe is accelerating.
- Predict the $z > 1$ deceleration observed by Riess et al. [2004]: the DNN trained on $z < 1$ data would extrapolate the accelerating trend, not the decelerating one, because it has no physical constraint requiring the cosmological transition.

Table 8 quantifies the contrast between the Friedmann model and a DNN at each epoch of the supernova program.

The photoelectric episode has the same structure. Einstein's quantum hypothesis reduced an infinite-dimensional function-learning problem to a two-parameter line-fitting problem. Friedmann's equations and the cosmological principle reduced the supernova distance-magnitude relation to a three-parameter curve, and Phillips reduced the scatter enough to make that curve measurable from ≈ 40 events. In both cases the theorists specified the smallest useful hypothesis class; the data then constrained the free parameters within it.

13 Conclusions

The supernova discovery of cosmic acceleration is, at its mathematical core, a three-parameter weighted nonlinear regression. The free parameters are $\mathcal{M}$ (a nuisance combination of absolute magnitude and Hubble constant), Ω_M (matter density), and Ω_Λ (cosmological constant / dark energy density). The measurement equation

$$m_{B,i} = \mathcal{M} + 5\log_{10}\tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda) + \varepsilon_i$$

Table 8: Sample-complexity comparison between the three-parameter Friedmann model (VC dimension = 3) and a three-layer DNN ($p = 593$ parameters, VC dim $\approx$ 5,700). "Sufficient n" for $\approx 1\sigma$ determination of q_0 at 95% confidence is $O(\text{VC dim}/\epsilon^2)$ for the DNN; 3 for the Friedmann model (one more than the number of free parameters).

Epoch	n available	Friedmann model	DNN (VC$\approx$ 5700)	DNN outcome
Pre-Phillips (before 1993)	~ 20 raw	Insufficient†	Insufficient	Cannot detect signal (noise > signal)
Calán/ Tololo 1996	18 (low-z)	Insufficient†	Insufficient	Cannot constrain curvature of $\mu(z)$
Discovery 1998	42–50	Sufficient	Insufficient ($\times 140$)	Overfits; no inference
JLA compilation 2011	~ 500	Sufficient	Insufficient ($\times 12$)	Partially learns shape; cannot extract q_0
Modern era 2020s	~ 2000	Sufficient	Borderline	Recovers $\mu(z)$ shape; cannot identify q_0

† Without high-z data, $(\Omega_M, \Omega_\Lambda)$ are not identified: only $\mathcal{M}$ can be constrained from low-z observations.

is derived directly from general relativity; the Friedmann equations are the physical hypothesis that specifies the functional form.

The central estimation challenge was not the regression itself but two other tasks: (i) *standardizing* the supernovae using the Phillips width–luminosity relation to reduce scatter from ≈ 0.4 mag to ≈ 0.15 mag (factor-of-2.7 improvement in signal-to- noise); and (ii) *assembling* high-z supernovae at $z = 0.3$–0.8, where the $\mu(z)$ curve of the accelerating model differs from that of the decelerating model by ≈ 0.2–0.3 mag.

The MLE results of both groups—$\Omega_M \approx 0.28$, $\Omega_\Lambda \approx 0.72$—give:

$$\begin{aligned}
\hat{q}_0 &= \frac{\hat{\Omega}_M}{2} - \hat{\Omega}_\Lambda = 0.14 - 0.72 = -0.58 \quad \text{(acceleration at } 3\sigma\text{)},\\
z_t &\approx 0.73 \quad \text{(deceleration–acceleration transition)},\\
\hat{\Omega}_\Lambda &= 0.72 \pm 0.09 \text{ (stat)} \quad \text{(dark energy confirmed by both teams)}.
\end{aligned}$$

Riess, Schmidt, and Perlmutter measured the curvature of the Hubble diagram and found $\Omega_\Lambda \approx 0.72$. They did not explain what dark energy is. The cosmological constant remains, in Weinberg's [1989] phrase, one of the deepest unsolved problems in physics.

A The Deceleration Parameter and the Second Friedmann Equation

The deceleration parameter q_0 follows from the second Friedmann equation. With matter (pressure $p_M = 0$) and cosmological constant ($p_\Lambda = -\rho_\Lambda c^2$):

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3} = -\frac{4\pi G \rho_M}{3} + \frac{\Lambda}{3}. \tag{34}$$

Evaluating at $t = t_0$ (today, $\rho_{M,0} = 3H_0^2\Omega_M/(8\pi G)$, $\Lambda = 3H_0^2\Omega_\Lambda$):

$$\frac{\ddot{a}_0}{a_0} = -\frac{H_0^2\Omega_M}{2} + H_0^2\Omega_\Lambda. \tag{35}$$

Therefore

$$q_0 = -\frac{\ddot{a}_0 a_0}{\dot{a}_0^2} = -\frac{\ddot{a}_0/a_0}{H_0^2} = \frac{\Omega_M}{2} - \Omega_\Lambda. \tag{36}$$

For $\Omega_M < 2\Omega_\Lambda$ the universe is accelerating today. The best-fit flat model ($\Omega_M = 0.28$, $\Omega_\Lambda = 0.72$) gives $q_0 = 0.14 - 0.72 = -0.58$: the universe currently accelerates at a rate $|q_0| = 0.58$ times the Hubble deceleration.

B Low-Redshift Expansion and the Hubble Diagram

At low redshifts, the luminosity distance has a Taylor expansion in z:

$$d_L(z) = \frac{cz}{H_0}\left[1 + \tfrac{1}{2}(1 - q_0)z + O(z^2)\right]. \tag{37}$$

The distance modulus then expands as

$$\mu(z) = 5\log_{10}\left(\frac{cz}{H_0\,\mathrm{Mpc}}\right) + 25 + \frac{5(1-q_0)}{2\ln 10}\,z + O(z^2). \tag{38}$$

At $z < 0.1$ (the Calán/Tololo range), the second term is $\approx 0.022\,(1 - q_0)z < 0.004$ mag, entirely negligible compared to $\sigma_{\text{int}} \approx 0.15$ mag. This confirms that the low-z sample constrains $\mathcal{M}$ through Hubble's law but cannot constrain q_0: the cosmological information is concentrated in the second-order term, which first becomes measurable for $z \gtrsim 0.3$.

The expansion also shows that the optimal experimental design places supernovae far into the nonlinear regime $z > 0.3$.

C The Low-Temperature Equation of State and Dark Energy

The cosmological constant Λ corresponds to a fluid with equation of state $w_\Lambda = p_\Lambda/(\rho_\Lambda c^2) = -1$: its pressure is equal and opposite to its energy density. A dark-energy fluid with $w \neq -1$ (e.g., a rolling scalar field ϕ with potential $V(\phi)$) also has $\ddot{a} > 0$ provided $w < -1/3$.

The most general measurement equation, allowing w as a free parameter, is

$$d_L(z; \Omega_M, \Omega_{\rm de}, w) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_M(1+z')^3 + \Omega_{\rm de}(1+z')^{3(1+w)}}}. \tag{39}$$

The 1998–1999 analyses assumed $w = -1$ (pure Λ); later surveys (SNLS, Union, JLA, DES) have extended the measurement to $w = -1.028 \pm 0.032$ [Conley et al., 2011, Sullivan et al., 2011], consistent with the cosmological constant at the 1% level. The discriminating power of current supernova samples now exceeds 100σ for $w < 0$ (dark energy required), with the remaining uncertainty in w lying in the determination of systematics rather than photon statistics.

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